

PATA

Done by

2300030941- Bhashya Palani Challagali

2400089009- T. Anusha

2300040075- A. Gowthami

Problem Statement

Indian addresses are often written as messy free text and may contain landmark-based directions, informal colony names, incorrect pincodes, spelling variations, or regional-language/transliterated text. This creates difficulties for last-mile delivery partners, who may spend significant time making phone calls, searching for locations, or navigating to inaccurate map pins.

The proposed system aims to solve this problem by taking a messy Indian address and converting it into a **single, accurately geocoded location**. It should resolve ambiguity by using nearby real-world landmarks from **OpenStreetMap** and validating the address against **pin code and locality reference data**, rather than simply guessing a location.

Abstract

The proposed **Location Intelligence System for Last-Mile Delivery** is an AI-based solution designed to accurately interpret and geocode messy Indian addresses. The system processes free-text addresses containing landmarks, informal locality names, incorrect pincodes, mixed scripts, and transliterated text.

It combines **address parsing, pincode reference data, and real-time OpenStreetMap landmark information** to identify and validate the most appropriate geographical location. The system also provides evidence for the resulting geocode, such as the matched pincode or nearby landmark, while recording automated corrections for auditability.

The solution is designed to provide fast and reliable geocoding suitable for use during order placement and last-mile delivery operations.

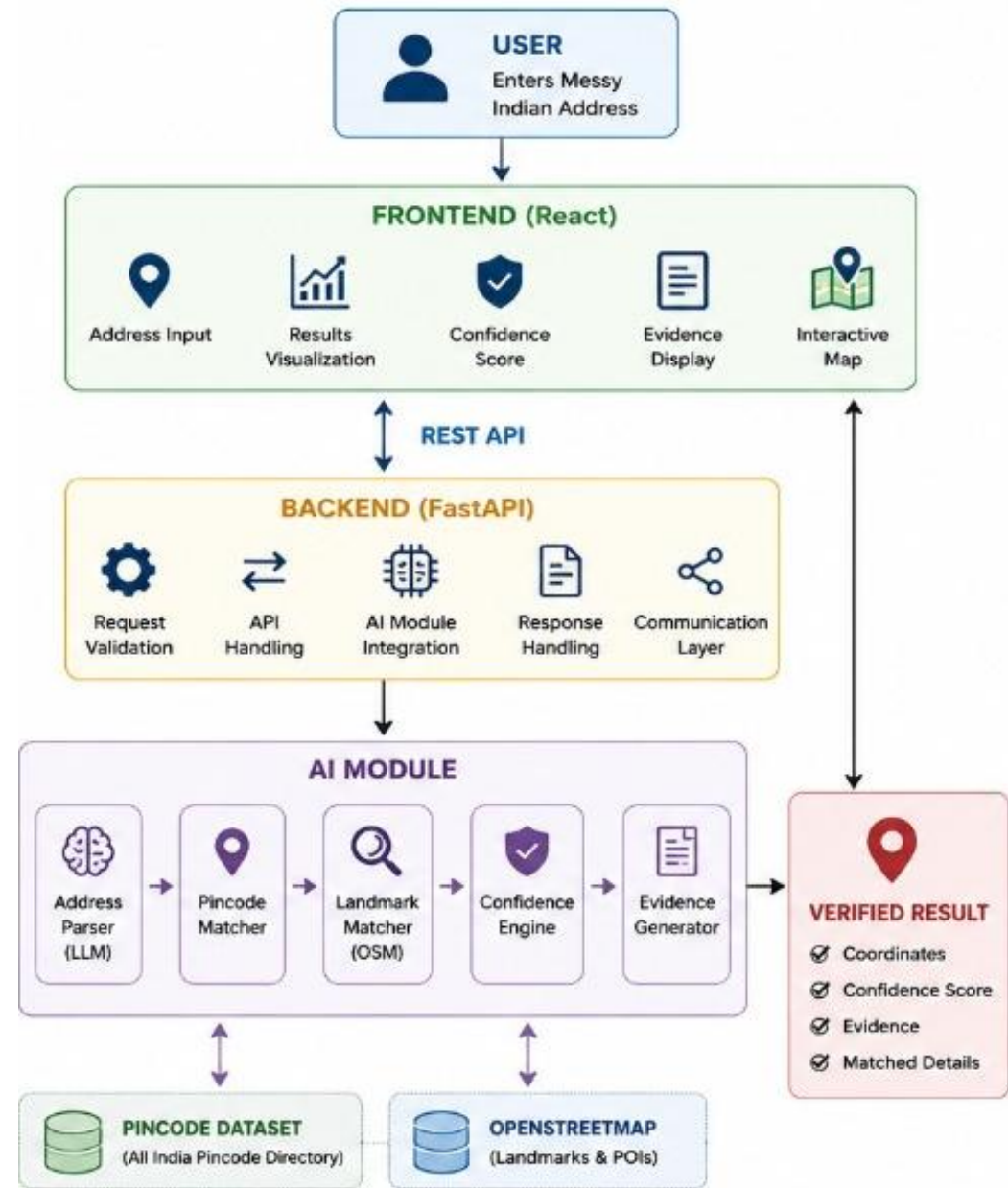
Objectives

- To parse **messy and unstructured Indian addresses**.
- To identify and validate **pincodes and localities** using reference data.
- To use **nearby landmarks from OpenStreetMap** to resolve address ambiguity.
- To convert an address into an **accurately geocoded latitude and longitude**.
- To identify and flag **low-confidence results** rather than silently guessing.
- To provide evidence explaining which **landmark or pincode match** contributed to the geocode.
- To provide results quickly enough for **order-placement-time usage**.
- To maintain an auditable record of automated address corrections while preserving the original address.

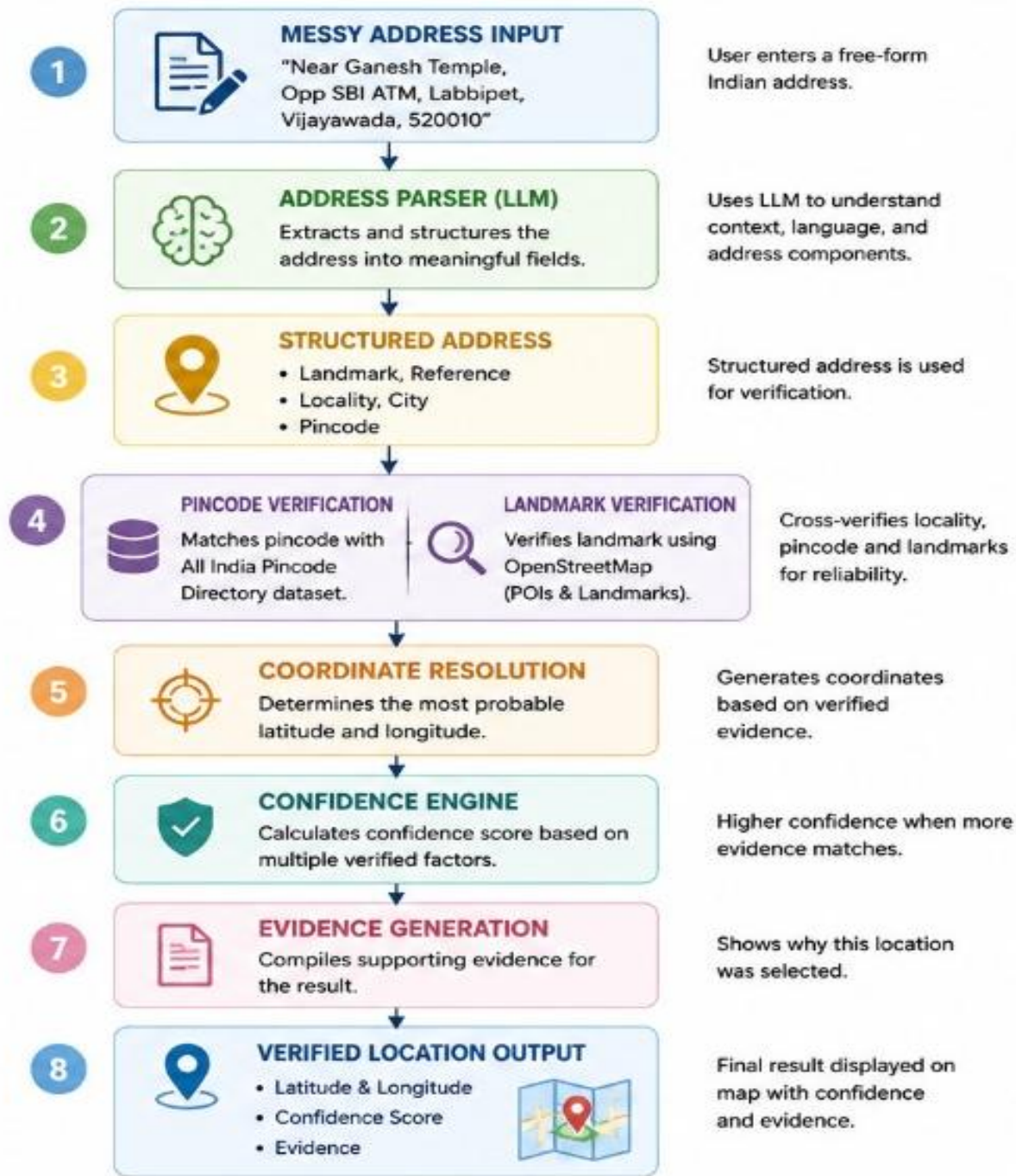
Introduction

- Indian addresses are often unclear and unstructured.
- Wrong pincodes and landmark-based directions can cause delivery delays.
- **Pata** uses AI, pincode data, and nearby landmarks to identify accurate locations.
- It helps improve the speed and accuracy of last-mile delivery.

System Architecture



AI workflow



Expected Results

User should enter the address

NAVIGATION

Dashboard

History (soon)

Batch Upload (soon)

API Docs

ABOUT

Enter any Indian address. We clean it, extract pincode, match landmarks and return coordinates with confidence.

ADDRESS INTELLIGENCE

Enter a messy Indian address and we'll locate it.

darsi,Ongole

 Find Location

CLEANED ADDRESS

darsi ongole

LATITUDE

15.314527

LONGITUDE

79.884888

CONFIDENCE SCORE



18%

Low

EVIDENCE

alternate match

NDC ONGOLE, PRAKASAM, ANDHRA PRADESH - 523003

MAP & DIRECTIONS

From: Your current location (16.44111, 80.62349)

To: darsi ongole

Distance: 183.7 km · **Est. time:** 2 h 23 min

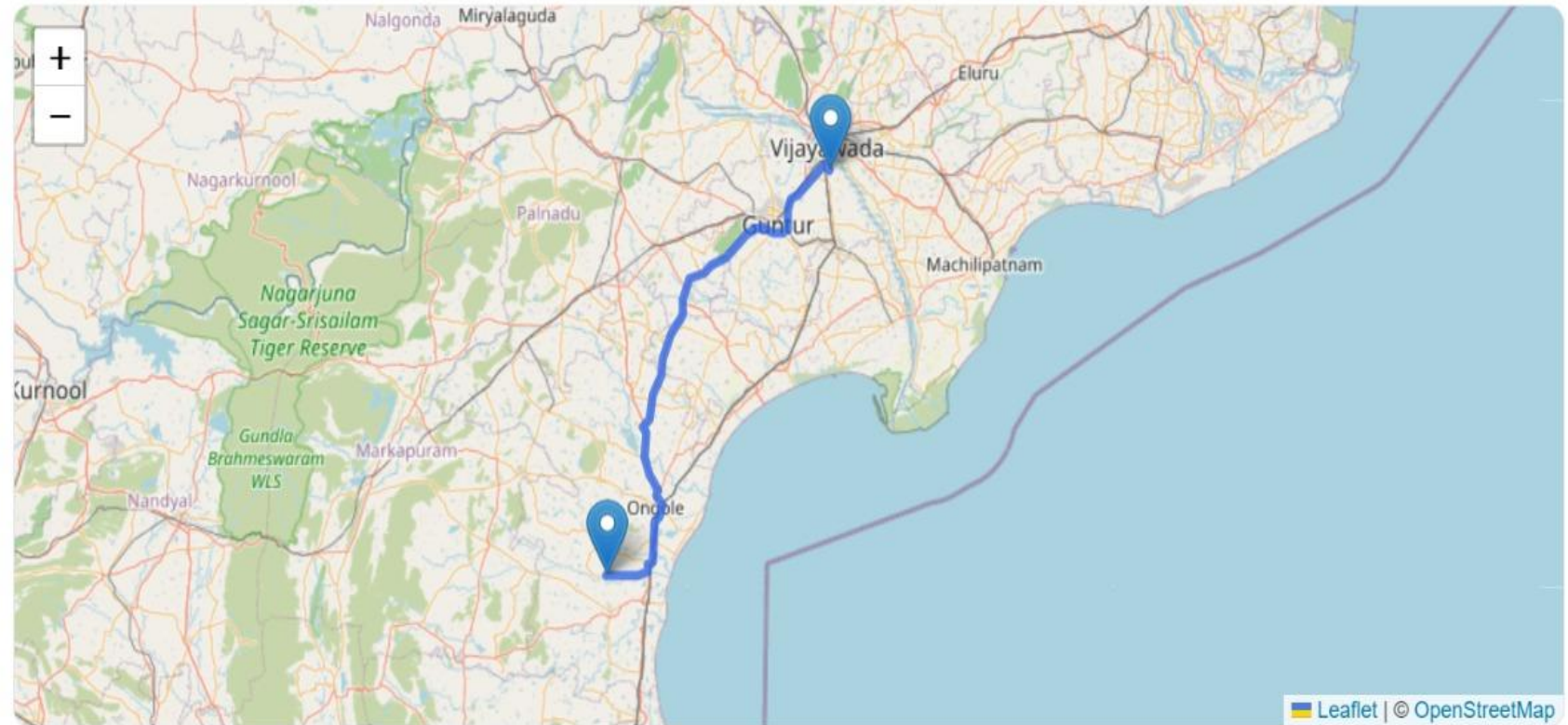
[Open directions in Google Maps](#)

From: Your current location (16.44111, 80.62349)

To: darsi ongole

Distance: 183.7 km · **Est. time:** 2 h 23 min

[Open directions in Google Maps](#)



NAVIGATION

Dashboard

History (soon)

Batch Upload (soon)

API Docs

ABOUT

Enter any Indian address. We clean it, extract pincode, match landmarks and return coordinates with confidence.

ADDRESS INTELLIGENCE

Enter a messy Indian address and we'll locate it.

Kl university.Guntur



Find Location

CLEANED ADDRESS

kl university guntur

LATITUDE

28.895583

LONGITUDE

76.592528

CONFIDENCE SCORE



clean it, extract pincode, match landmarks and return coordinates with confidence.

CLEANED ADDRESS

kl university guntur

LATITUDE

28.895583

LONGITUDE

76.592528

CONFIDENCE SCORE



12%

Low

EVIDENCE

alternate match

Central University of KarnatakaKad, KALABURAGI, KARNATAKA - 585367

alternate match

Central University Haryana Mohindergarh SO, MAHENDRAGARH, HARYANA - 123031

alternate match

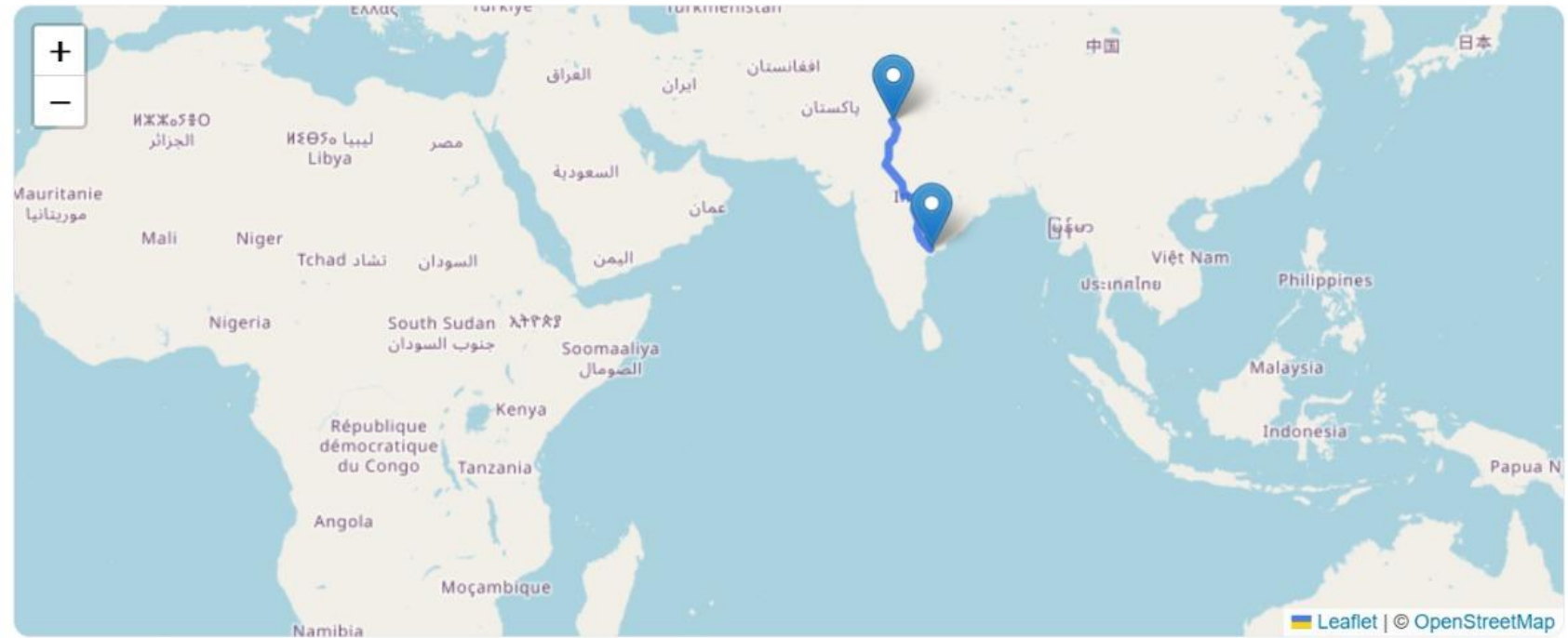
Central University of Rajasthan S.O, AJMER, RAJASTHAN - 305817

From: Your current location (16.44111, 80.62349)

To: kl university guntur

Distance: 1875.4 km · **Est. time:** 22 h 36 min

[Open directions in Google Maps](#)



THANK YOU